Survey and Analysis of

Quantum Processing Integration with Large Language Models (LLMs)

MBA Project Report

Vigneshwara Chinnadurai
Roll No: 2414504298 | Elective: Analytics & Data Science
Guide: Mr. Govind

Centre for Distance and Online Education | Manipal University Jaipur | May 2026

Introduction & Background

The Challenge:

- LLMs (GPT-4, Gemini) require enormous computational resources
- GPT-4 training estimated at $100M+ in compute costs
- Exponential growth in model parameters (1.5B → 1.8T in 4 years)

The Opportunity:

- Quantum computing offers fundamentally different computation
- Superposition: n qubits represent 2^n states simultaneously
- Quantum parallelism could accelerate attention mechanisms
- Potential for exponential speedup in specific sub-tasks

Research Question:

- How can quantum computing be integrated with LLMs?
- What is the current state of Quantum NLP research?

Research Objectives

- To comprehensively review and synthesize research on quantum computing integration with LLMs (2017-2025)

- To analyze and categorize existing approaches: quantum-inspired, hybrid architectures, prototype QNLP models

- To summarize technology trends and barriers to adoption

- To conduct hands-on experimentation with quantum simulators (IBM Qiskit, PennyLane, Google Cirq)

- To provide strategic recommendations for future research and practical integration in analytics/data science

Research Methodology

Mixed-Methods Approach:

1. Systematic Literature Review

- 47 papers analyzed from arXiv, IEEE, Google Scholar (2017-2025)
- Thematic coding & technology maturity assessment

2. Experimental Research (4 Experiments)

- Exp 1: Quantum Word Encoding (Amplitude, Angle, IQP)
- Exp 2: Quantum Text Classification (Variational Circuits)
- Exp 3: Hybrid Quantum-Classical Pipeline Comparison
- Exp 4: Noise Analysis & Cross-Framework Benchmarking

Tools Used:

- IBM Qiskit | PennyLane | Google Cirq | scikit-learn | Python

Literature Review: Key Findings

Publication Growth:

- 3 papers (2017-18) → 21 papers (2023-25): exponential growth

Research Distribution:

- Theoretical/Framework: 29.8%
- Simulation-Only: 38.3%
- Hardware-Validated: 17.0%

Key Developments:

- DisCoCat framework (Coecke et al., 2020): NLP → Quantum circuits
- Quantum Transformers (Beer et al., 2021): Quantum attention
- lambeq library: Practical QNLP toolkit
- Quantinuum H2: First hardware-validated sentence classification

Key Experimental Results

Experiment 1: Quantum Word Encoding

- 94.2% encoding fidelity | 8.3:1 compression ratio (50d → 6 qubits)

Experiment 2: Quantum Text Classification

- 87.3% accuracy with only 48 parameters (vs ~3000 for classical NN)
- Competitive with SVM and Logistic Regression baselines

Experiment 3: Hybrid Quantum-Classical Pipeline

- +8.5% accuracy advantage in low-data regime (n=50 samples)
- 133x parameter reduction vs classical neural network

Experiment 4: Noise Resilience & Benchmarking

- -5.8% accuracy drop at realistic noise level (p=0.01)
- Hybrid approaches assessed at TRL 5-6 maturity

Conclusions

1. Quantum advantages are real but specific:

- Encoding compression (94.2% fidelity, 8.3:1 ratio)
- Small-data learning (+8.5% accuracy at n=50)
- Parameter efficiency (133x reduction)

2. Hybrid approaches are the practical path forward

- Best overall score in multi-criteria evaluation (27/35)
- Competitive accuracy with dramatic parameter savings

3. Timeline: 10-15 years for full quantum LLMs

- Near-term (1-3 yrs): Hybrid pilots on NISQ hardware
- Medium-term (3-7 yrs): Quantum-enhanced sub-routines
- Long-term (7-15 yrs): Fault-tolerant quantum transformers

Recommendations

For Organizations:

- Establish quantum literacy programs for data science teams
- Identify NLP use cases with small-data characteristics
- Launch hybrid pilot projects using cloud quantum platforms
- Develop 5-year quantum readiness roadmaps

For Researchers:

- Develop standardized QNLP benchmarks
- Focus on noise-resilient algorithms for NISQ hardware
- Explore quantum advantages for low-resource languages

Phased Adoption Framework:

- Phase 1 (2025-27): Literacy + Simulators
- Phase 2 (2027-30): Hybrid Pilots on NISQ
- Phase 3 (2030+): Production Quantum-Enhanced NLP

Thank You

Questions & Discussion

Vigneshwara Chinnadurai
Roll No: 2414504298 | MBA - Analytics & Data Science

Centre for Distance and Online Education | Manipal University Jaipur